

INTRO TO ITWS — QUIZ 1 MASTER CRIB SHEET

QUICK LOOKUP INDEX

- TCP Handshake → SYN / SYN-ACK / ACK
- DNS → Domain name to IP address resolution
- TCP vs UDP → Reliable vs Fast
- CSS Specificity → Inline > ID > Class > Element
- HTML vs XHTML → Forgiving vs Strict
- Block vs Inline → Layout vs Flow
- XML Well-Formed → One root, closed tags, proper nesting
- Web Science → Web + society as a system
- Generative vs Agentic AI → Responds vs Acts

LECTURE 1 — INTRO TO ITWS & WEB SCIENCE

Core Concepts

ITWS studies how technology, people, and information systems interact at scale. The web is a socio-technical system combining technical infrastructure with human behavior.

Key Vocabulary

Web — Global system of interlinked documents accessed via HTTP/HTTPS.

Internet — Network of networks using TCP/IP.

Socio-technical system — Interaction of technology and people.

Scalability — Ability to grow without failure.

Emergent behavior — Outcomes from many interactions.

How This Appears on the Quiz

Web Science matters because it allows us to analyze and improve large-scale web systems by considering both technical infrastructure and human behavior.

LECTURE 2 — NETWORKS, PROTOCOLS, TCP/IP

Core Concepts

Internet communication occurs in layers, each with a specific responsibility. Data is encapsulated as it moves through the stack.

Key Vocabulary

Protocol — Rules for data communication.

Packet — Unit of transmitted data.

Latency — Transmission delay.

Bandwidth — Data capacity.

TCP Three-Way Handshake

```
SYN      -> Client requests connection
SYN-ACK  <- Server acknowledges
ACK      -> Client confirms
```

What this does: Establishes a reliable connection so both client and server are ready to send and receive data.

How This Appears on the Quiz

DNS translates domain names to IP addresses, enabling computers to locate servers. TCP ensures reliable communication before HTTP data transfer.

LECTURE 3 — HTML, XHTML, HTML5

Core Concepts

HTML provides semantic structure, while CSS controls presentation. HTML5 adds meaning and native support for modern web features.

Annotated HTML Example

```
<!DOCTYPE html>          <!-- Declares HTML5 -->
<html>                   <!-- Root element -->
  <head>                  <!-- Metadata -->
    <title>Page</title><!-- Browser tab -->
  </head>
  <body>                  <!-- Visible content -->
    <p>Hello</p>          <!-- Paragraph -->
  </body>
</html>
```

What this does: Defines the structure of an HTML5 document so browsers correctly render and interpret content.

LECTURE 4 — CSS & SELECTORS

Core Concepts

CSS separates presentation from content. When multiple rules apply, specificity determines which rule wins.

CSS Example

```
p { color: red; }          /* Element selector */
#title { color: blue; }    /* ID selector */
```

What this does: Demonstrates how more specific selectors override general ones.

LECTURE 5 — XML & STRUCTURED DATA

Core Concepts

XML stores and transports data using strict, user-defined tags.

```
<?xml version="1.0"?>
<book>
  <title>Example</title>
</book>
```

What this does: Stores structured data in a machine-readable format.

LECTURE 6 — AI: GENERATIVE VS AGENTIC

Core Concepts

Generative AI responds to prompts, while agentic AI acts autonomously toward goals.

LABS — WEBSITE & VM DEPLOYMENT

```
sudo chown -R www-data:www-data /var/www/html
# Gives Apache ownership of site files
```

What this does: Allows the web server to read and serve website files securely.

CASE DISCUSSION — KUEHCHIC DESSERTS

Key risks include task delays on the critical path and limited manpower. Time is the most critical constraint.